Replicating LEAR on MIBEL through the Solar Transition: Why Technical Indicators Do Not Transfer, and Why Percentage Errors Mislead (2022–2024)

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Abstract

We replicate the LEAR day-ahead electricity-price forecaster (Lago et al., 2021) on the Iberian market (MIBEL) over 2022–2024 — a window that spans both the gas-crisis price regime and the onset of structural solar saturation — and report two cautionary findings that the replication makes visible. The backbone result is reassuring: LEAR, recalibrated daily, beats a seasonal-naive benchmark in every regime (relative MAE 0.45–0.57; Diebold–Mariano $p < 0.001$ throughout), with a demand-plus-wind exogenous block the best variant. But two negative results are the contribution. First, the technical indicators of Demir et al. (2020), helpful in their original setting, exhibit *negative transfer* to MIBEL: adding them degrades accuracy in every regime (relative MAE worse by 4.8–9.5 points), because momentum/trend features add variance, not signal, in a market whose price is already pinned by the demand and wind covariates LEAR carries. Second, and more general, error *metrics* diverge sharply as solar penetration rises: the share of near-zero-price hours grows from 0.3% (2022) to 12.4% (2024), and over the same span sMAPE explodes from 16% to 51% while the mean absolute error actually *falls* (18.3 to 14.0 €/MWh). The same model with the same skill thus looks three times worse or somewhat better depending only on the metric, because percentage errors blow up as the denominator vanishes (Hyndman and Koehler, 2006). We draw a concrete recommendation for forecasting high-renewable markets and frame both findings as transferable cautions rather than a claim to a better forecaster.

Keywords: electricity price forecasting; LEAR; technical indicators; forecast accuracy metrics; solar saturation; MIBEL. **JEL:** C53, Q47, C52.

1 Introduction

The LEAR model (Lasso-Estimated AutoRegressive) is a standard, strong open benchmark for day-ahead electricity-price forecasting (Lago et al., 2021). We port it faithfully to MIBEL and evaluate it across 2022–2024, a period that is unusually informative because it contains two distinct stresses: the 2022 gas-crisis price regime and, by 2024, the structural solar saturation that drives Iberian day-ahead prices to zero for an increasing fraction of hours. A faithful replication on this period is useful in itself, but its real value here is that it surfaces two cautions that generalise beyond MIBEL. We are explicit that this is a replication-plus-diagnosis study, not a new forecaster: the contribution is the two negative results, which are robust, grounded, and directly useful to practitioners.

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Table 1: LEAR (demand+wind) vs. seasonal naive, by regime. rMAE < 1 means LEAR beats naive; Diebold–Mariano p against naive.

Regime	Test hours	MAE (€/MWh)	sMAPE (%)	rMAE
2022 (H2, crisis)	4,272	18.3	16.1	0.566
2023 (full year)	8,663	12.9	24.6	0.452
2024 (full year)	8,591	14.0	51.0	0.511
2023–2024 pooled	17,254	13.5	37.8	0.481
Diebold–Mariano $p < 0.001$ in every row.				

2 Data and method

Prices are the OMIE day-ahead Spanish series (ESIOS indicator 600). The exogenous regressors are ESIOS *forecast* (*previsión*) products — day-ahead demand (indicator 460), wind (541) and solar (540) — which are forecasts by construction, not realised series, and against which the target is masked. We report an AR-only configuration (no exogenous data at all) and a demand+wind configuration; we flag in §6 a residual caveat on the *vintage* of the ESIOS forecasts. LEAR follows Lago et al. (2021): a Lasso over calendar dummies and price/exogenous lags, recalibrated each day on a rolling window; the benchmark is a seasonal-naive predictor (same hour, previous similar day). Skill is the relative MAE (rMAE = model MAE / naive MAE) and significance the Diebold and Mariano (1995) test with a Newey–West correction (lag 23). Following Lago et al. (2021) the symmetric MAPE is the bounded form

$$\text{sMAPE} = \frac{100}{n} \sum_t \frac{2\,|y_t - \hat{y}_t|}{|y_t| + |\hat{y}_t|} \in [0, 200],$$

in which the only undefined points are the measure-zero set with $|y_t| + |\hat{y}_t| = 0$ (both exactly zero); we drop exactly those, and since LEAR's forecast is essentially never exactly zero, no hours are effectively removed — the metric is computed over the full test set. We evaluate three regimes — 2022 H2 (crisis), 2023 and 2024 (full years) — plus the 2023–2024 pool, on 4,272–17,254 test hours each.

3 Result 1: LEAR is robust across regimes (the backbone)

Table 1 reports LEAR with the demand-plus-wind exogenous block, the best-performing variant. It beats the naive benchmark decisively in every regime (rMAE 0.45–0.57, all $p < 0.001$), including the crisis. Adding a solar forecast on top of demand and wind does *not* help (rMAE marginally worse in 2023–2024), consistent with the wind and demand forecasts already spanning most of the predictable variation. This is a clean, reproducible replication of LEAR's known strength; it is the platform for the two findings below, not the contribution.

4 Result 2: technical indicators do not transfer

Demir et al. (2020) introduced financial technical indicators (EMA, Bollinger bands, MACD, momentum, ROC, Coppock) as engineered features for electricity-price forecasting, reporting gains in their setting. On MIBEL they exhibit clear *negative transfer*. Table 2 adds the same indicators to the demand+wind LEAR: accuracy degrades in *every* regime, by 4.8–9.5 points of rMAE, and the degradation is largest in the calmer, solar-heavy 2023–2024 years. The Lasso prunes almost all of the indicator coefficients, yet those it retains add variance rather than signal:

Table 2: Technical-indicator transfer: relative MAE of LEAR (demand+wind) without vs. with the Demir et al. (2020) indicators. Positive Δ = accuracy lost.

Regime	rMAE (base)	rMAE (+TI)	Δ (pts)
2022 (H2, crisis)	0.566	0.615	+4.8
2023 (full year)	0.452	0.538	+8.6
2024 (full year)	0.511	0.605	+9.5
2023–2024 pooled	0.481	0.565	+8.4

in a market whose day-ahead price is already pinned by demand, wind and its own autoregressive structure, momentum/trend features — designed for slower, expectations-driven financial series — have little incremental information and hurt out-of-sample. The practical reading is that technical indicators should not be imported into high-renewable EPF without a market-specific transfer test.

5 Result 3: metrics diverge under solar saturation

The most general finding concerns measurement. As Iberian solar capacity grows, the day-ahead price hits (near-)zero for a rising share of hours: from 0.3% of hours at or below 1 €/MWh in 2022 to 2.0% in 2023 and 12.4% in 2024 (8.9% are exactly zero or negative). The bounded symmetric MAPE we use (§2) does *not* divide by zero — the $|y_t| + |\hat{y}_t| = 0$ set is dropped and is empty in practice — but that guard is exactly why the metric misleads rather than crashes. Each near-zero hour, where $|y_t| + |\hat{y}_t|$ is small, contributes a term near the 200% ceiling almost regardless of the absolute error, so the *average* sMAPE rises mechanically with the *count* of near-zero hours, not with forecast error (Hyndman and Koehler, 2006). Figure 1 and Table 1 show the consequence — holding the model fixed (LEAR demand+wind), sMAPE rises from 16% (2022) to 25% (2023) to 51% (2024) in lockstep with the near-zero share, while the mean absolute error over the same span *falls* from 18.3 to 14.0 €/MWh and the relative MAE is essentially flat ($\sim$0.5). The same forecaster, with the same underlying skill, looks three times worse or modestly better depending only on whether one reads sMAPE or MAE.

The recommendation is concrete: in high-renewable markets, and for any comparison across years or markets with different solar penetration, accuracy must be reported in absolute terms (MAE) and relative to a benchmark (rMAE), and percentage metrics (MAPE, sMAPE) either dropped or restricted to non-zero-price hours with the exclusion disclosed. A literature that ranks forecasters or markets on sMAPE is, increasingly, ranking them on their solar share.

6 Limitations and conclusion

This is a replication and diagnosis, not a new model: we do not implement the deep neural benchmark of Lago et al. (2021) nor extend to the continuous intraday market, both of which we leave for future work.

A sharper caveat concerns the *vintage* of the exogenous forecasts. The demand and wind regressors are ESIOS *previsión* (forecast) products, not realised series, so the gross form of leakage — feeding the model the realised wind and demand it is implicitly trying to price — is avoided by construction. But a historical pull made in 2026 returns whatever value ESIOS currently stores for each past timestamp, and we have *not* independently confirmed that this is the original $D-1$ forecast frozen at publication rather than a later revision. If REE revised those forecast series ex post toward the realised outturn, the demand+wind results would carry

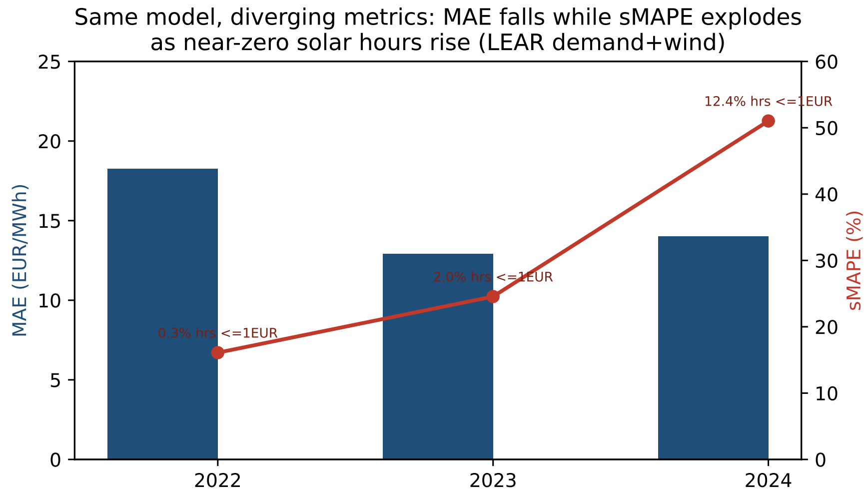

Figure 1: Same model (LEAR demand+wind), diverging metrics: as the share of near-zero-price hours rises (annotated), sMAPE explodes while MAE falls.

an optimistic look-ahead bias and the Diebold–Mariano significance on that increment would be overstated. We flag this as the first verification a referee should demand (pulling each indicator's publication timestamp per calendar day) and report it honestly rather than asserting a clean audit. Crucially, the backbone result does *not* rest on these series: the **AR-only** LEAR, which consumes *no* exogenous data and is therefore immune to any vintage concern, also beats the naive benchmark in every regime (rMAE 0.52–0.62, Diebold–Mariano $p < 0.001$). The demand+wind block only sharpens an already-significant margin.

Within that scope the conclusions are firm. LEAR transfers cleanly to MIBEL and is robust across the crisis and the solar-transition regimes. Two cautions, however, travel with that result and beyond it: the technical indicators of Demir et al. (2020) exhibit negative transfer and should not be adopted without a market-specific test, and percentage error metrics become unreliable exactly as a market decarbonises, so high-renewable EPF should be scored on absolute and relative errors. The negative results, not the replication, are the contribution.

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